Problem Set 5

Due May 27, 4:30 pm

1. Peng Ding's book

- Chapter 21: 21.1, 21.6, 21.7 (data in Rain classroom)
- Chapter 22: 22.2, 22.7
- Chapter 23: 23.1, 23.2 (optional), 23.3

2. Wald and two-stage least squares

We derived the identification formula for CACE under randomization in class. Show that

$$\text{CACE} = \frac{\text{cov}(Z_i, Y_i)}{\text{cov}(Z_i, D_i)}.$$

3. Weighting method for CACE in observational studies (optional)

Suppose that the randomization assumption of Z_i is replaced by ignorability $Z_i \perp\!\!\!\perp (Y_i(1), Y_i(0), D_i(1), D_i(0)) \mid X_i$, while monotonicity and the exclusion restriction continue to hold. Define

$$\begin{aligned} \kappa_{0i} &= (1 - D_i)\frac{(1 - Z_i) - \text{pr}(Z_i = 0 \mid \boldsymbol{X}_i)}{\text{pr}(Z_i = 0 \mid \boldsymbol{X}_i)\text{pr}(Z_i = 1 \mid \boldsymbol{X}_i)}, \\ \kappa_{1i} &= D_i\frac{Z_i - \text{pr}(Z_i = 1 \mid \boldsymbol{X}_i)}{\text{pr}(Z_i = 0 \mid \boldsymbol{X}_i)\text{pr}(Z_i = 1 \mid \boldsymbol{X}_i)}. \end{aligned}$$

Show that

$$
\begin{aligned}
\mathbb{E}\{g(Y_i(0), \boldsymbol{X}_i) \mid U_i = c\} &= \frac{1}{\mathrm{pr}\{U_i = c\}} \mathbb{E}\{\kappa_{0i} g(Y, X)\}, \\
\mathbb{E}\{g(Y_i(1), \boldsymbol{X}_i) \mid U_i = c\} &= \frac{1}{\mathrm{pr}\{U_i = c\}} \mathbb{E}\{\kappa_{1i} g(Y, X)\}.
\end{aligned}
$$

4. Analysis of return to schooling

Card (1993) used the geographic variation in college proximity to estimate the return to schooling. The treatment is education and the outcome is log wage. The instrumental variable is the college proximity. The data file *card.csv* is a slightly modified version of the data from Card's homepage. The explanations of the variable names are in *code_bk.txt* (column location information is not correct and some variables are not included in the data provided). When you do the data analysis, please give details about your causal quantities, assumptions, model choices, and interpret your results. You can read Card (1993) for more details about the data and suggestions about model specification.

(a) You can dichotomize the education variable and infer the CACE (the effect for units with $D(1) = 1, D(0) = 0$ where D is the dichotomized education) without covariates. How do you choose the threshold for the dichotomization? Are the results sensitive to your choice? Conduct an analysis with covariates.

(b) Without dichotomizing the education variable, you can use the standard two-stage least squares estimator. Compare the results to the analysis above.

REFERENCES

Card, D. (1993). Using geographic variation in college proximity to estimate the return to schooling. Technical report, National Bureau of Economic Research.